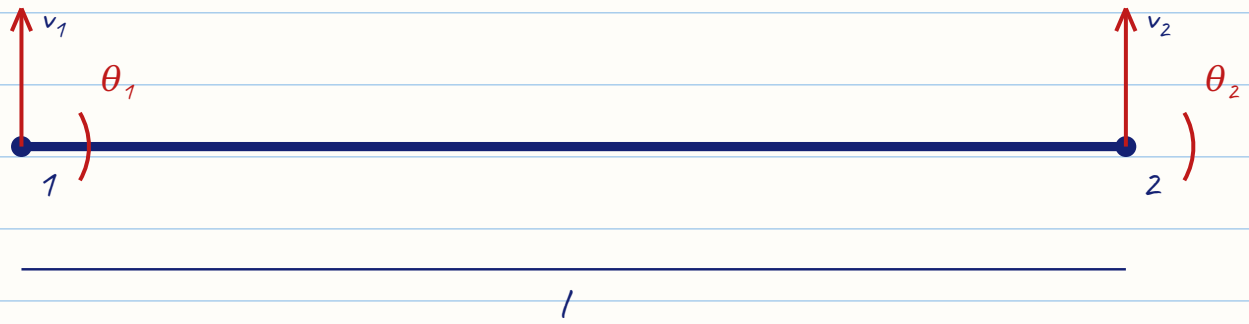
# 1. Introduction to Beam Elements

## 1.1 Beam element

- A beam is a long slender member that carries transverse loads and resists them by bending.
- Theory used: Euler-Bernoulli beam theory – plane sections remain plane and perpendicular to the neutral axis, shear deformation is ignored.
- Each node of the beam element has 2 degrees of freedom: transverse deflection v and slope (rotation) $\theta = dv/dx$.
- Nodal forces correspond to shear force F and bending moment M.




EI = constant; 2 dof per node $(v, \theta) \rightarrow$ 4 dof per element

## 1.2 Assumptions

- Material is linearly elastic, homogeneous and isotropic.
- Cross-section is constant along the element; EI is constant.
- Deflections are small; slope dv/dx is small.
- Loads act in the plane of symmetry; loads are applied only at nodes

(equivalent nodal loads for UDL).

### 1.3 Basic beam relations

$M = EI\, d^2v/dx^2$, curvature $\kappa = d^2v/dx^2$

Shear force $F = dM/dx$, Load $w = dF/dx$

Bending strain $\varepsilon = -y\, d^2v/dx^2$, $\sigma = E\varepsilon = -E\, y\, d^2v/dx^2$

Governing equation: $EI\, d^4v/dx^4 = w(x)$

## 2. Displacement Function for the Beam Element

### 2.1 Cubic polynomial

- Four nodal dof $(v_1, \theta_1, v_2, \theta_2)$ → four constants → cubic displacement:

$$v(x) = a_1 + a_2x + a_3x^2 + a_4x^3$$

$$\theta(x) = dv/dx = a_2 + 2a_3x + 3a_4x^2$$

- Boundary conditions at the two nodes:

at $x = 0$: $v = v_1$, $\theta = \theta_1$

at $x = l$: $v = v_2$, $\theta = \theta_2$

### 2.2 Solving for constants

$$a_1 = v_1,\ a_2 = \theta_1$$

$$a_3 = -3(v_1 - v_2)/l^2 - (2\theta_1 + \theta_2)/l$$

$$a_4 = 2(v_1 - v_2)/l^3 + (\theta_1 + \theta_2)/l^2$$

# 3. Shape Functions of the Beam Element

- Substitute the constants and rearrange: $v = [N]\{q\}$, $\{q\} = \{v_1\ \theta_1\ v_2\ \theta_2\}^T$.

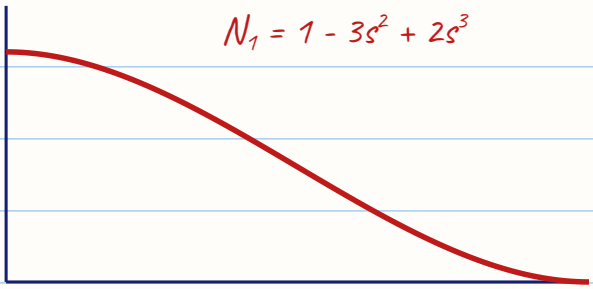
$$N_1 = 1 - 3x^2/l^2 + 2x^3/l^3$$

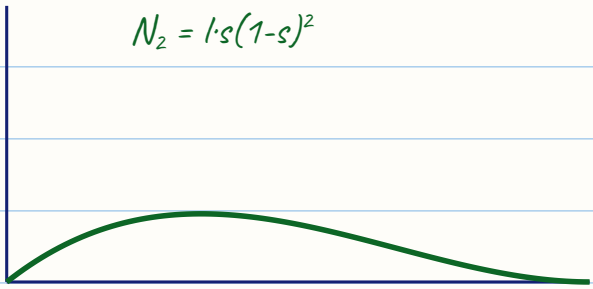
$$N_2 = x - 2x^2/l + x^3/l^2$$

$$N_3 = 3x^2/l^2 - 2x^3/l^3$$

$$N_4 = -x^2/l + x^3/l^2$$

- These are the Hermite shape functions ($C^1$ continuity).
- Properties: $N_1(0) = 1$, $N_3(l) = 1$; $dN_2/dx = 1$ at $x = 0$; $dN_4/dx = 1$ at $x = l$; $N_1 + N_3 = 1$.




$s = x / l$ (Hermite cubics)

# 4. Stiffness Matrix of the Beam Element

## 4.1 Strain–displacement matrix

- Curvature $d^2v/dx^2 = [B]\{q\}$ where $[B] = d^2[N]/dx^2$:

$$[B] = [\,-6/l^2 + 12x/l^3,\ -4/l + 6x/l^2,\ 6/l^2 - 12x/l^3,\ -2/l + 6x/l^2\,]$$

## 4.2 Strain energy approach

- Strain energy of the beam element:

$$U = \tfrac{1}{2} \int_0^l EI \left(d^2v/dx^2\right)^2 dx = \tfrac{1}{2} \{q\}^T [k] \{q\}$$

$$[k] = \int_0^l EI\, [B]^T [B]\, dx$$

## 4.3 Result

$$[k] = (EI/l^3) \begin{bmatrix} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{bmatrix}$$

- Order of dof: $v_1, \theta_1, v_2, \theta_2$. Corresponding forces: $F_1, M_1, F_2, M_2$.

$$\{F_1\ M_1\ F_2\ M_2\}^T = [k] \{v_1\ \theta_1\ v_2\ \theta_2\}^T$$

- Properties: symmetric, singular (rigid body motion possible), each column is a set of equilibrium forces for a unit displacement.

## 4.4 Physical meaning of the terms

- Column 1: unit deflection at node 1 ($v_1=1$, others 0) needs forces $12EI/l^3$, $6EI/l^2$, $-12EI/l^3$, $6EI/l^2$.
- Term $4EI/l$ = moment needed for unit rotation at one node with the other end fixed; $2EI/l$ = carry-over moment.
- Term $12EI/l^3$ = force required for unit deflection with no rotation at either end.

Remember: $EI/l^3$ is common factor; the matrix is 4×4 for one beam element.

# 5. Equivalent Nodal Loads (Load Vector)

## 5.1 Consistent load vector

- A distributed load $w(x)$ is replaced by equivalent nodal forces and moments:

$$\{f\} = \int_0^l [N]^T w(x)\, dx$$

## 5.2 Uniformly distributed load w over the whole element

$$\{f\} = \{\, wl/2 \;\; wl^2/12 \;\; wl/2 \;\; -wl^2/12 \,\}^T$$

## 5.3 Point load P at distance a from node 1 (b = l - a)

$$\{f\} = \{\, Pb^2(l+2a)/l^3 \;\; Pab^2/l^2 \;\; Pa^2(l+2b)/l^3 \;\; -Pa^2b/l^2 \,\}^T$$

- For a central load P: $\{f\} = \{\, P/2 \;\; Pl/8 \;\; P/2 \;\; -Pl/8 \,\}^T$.

## 5.4 Linearly varying (triangular) load, zero at node 1 and $w_0$ at node 2

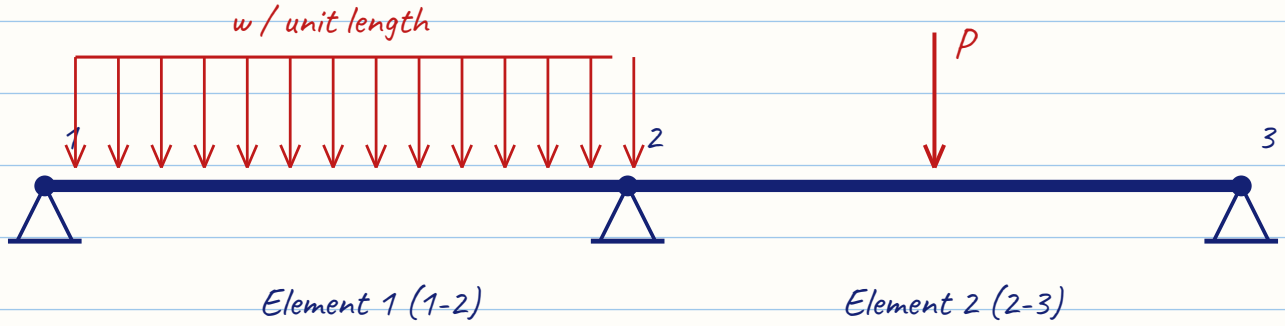
$$\{f\} = \{\, 3w_0l/20 \;\; w_0l^2/30 \;\; 7w_0l/20 \;\; -w_0l^2/20 \,\}^T$$

The fixed-end reactions of the same load have the opposite sign – they are the "fixed end forces".

# 6. Analysis of Continuous Beams

## 6.1 Continuous beam

- A beam resting on more than two supports. It is statically indeterminate; FEM treats it directly without extra compatibility equations.
- Nodes are placed at supports, at load points and at changes of section / material.




Continuous beam: elements joined at the support nodes

## 6.2 Procedure

- Step 1: Number the nodes (supports, load points) and elements. Each node has 2 dof (v, θ); total dof = 2 × number of nodes.
- Step 2: Find the stiffness matrix [k] of each element (4×4) using its own EI and l.
- Step 3: Find the equivalent nodal load vector of each element and combine it with applied point loads/moments.
- Step 4: Assemble the global stiffness matrix [K] by adding overlapping terms at common nodes (dof of shared node are the same for both elements).
- Step 5: Write $[K]\{U\} = \{F\}$.
- Step 6: Apply boundary conditions: deflection v = 0 at every support; slope θ = 0 at a fixed end. Eliminate those rows and columns.
- Step 7: Solve for the remaining unknown dof (slopes at supports, deflections at free points).
- Step 8: Find the support reactions and the element end forces:

$$\{f^{(e)}\} = [k^{(e)}]\{q^{(e)}\} - \{f_{equivalent}\}$$

- Step 9: Draw the shear force and bending moment diagrams.

## 6.3 Assembly for two elements

- Two elements (1-2) and (2-3), nodes 1, 2, 3; dof order $v_1\ \theta_1\ v_2\ \theta_2\ v_3\ \theta_3$.

  [K] is 6×6. Terms of the shared node 2 are added:

  $K_{33} = (12EI/l^3)_1 + (12EI/l^3)_2$, $K_{34} = (-6EI/l^2)_1 + (6EI/l^2)_2$

  $K_{44} = (4EI/l)_1 + (4EI/l)_2$

# 7. Solved Example – Propped / Two-span Beam

- Problem: A continuous beam ABC, simply supported at A and C and at B. Span AB = BC = L, EI constant, and carries a UDL of w per unit length on both spans. Find the slopes at the supports.
- Elements: (1) A–B, (2) B–C. dof: $v_A = v_B = v_C = 0$ (supports), slopes $\theta_A$, $\theta_B$, $\theta_C$ unknown.

  Reduced equations (rows/cols of θ only):

  $(EI/L)\ [\,4\ 2\ 0\,;\,2\ 8\ 2\,;\,0\ 2\ 4\,]\ \{\theta_A\ \theta_B\ \theta_C\}^T = \{\,wL^2/12\ 0\ -wL^2/12\,\}^T$

- By symmetry $\theta_B = 0$ (middle support):

  $(EI/L)(4\theta_A + 2\theta_B) = wL^2/12 \rightarrow \theta_A = wL^3/48EI$

  $\theta_C = -wL^3/48EI$, $\theta_B = 0$

- Support moment at B from element 1 ($\theta_B = 0$):

  $M_B = (2EI/L)\,\theta_A + wL^2/12 = wL^2/24 + wL^2/12 = wL^2/8$ (hogging)

- This matches the result of the three-moment equation.

Always check: sum of reactions = total load, and the moment at the end simple supports = 0.

## 8. Beam with Axial Load – Frame Element (Short Note)

- When axial deformation is also present, a plane frame element has 3 dof per node (u, v, θ) → 6×6 stiffness matrix: the bar terms AE/l are combined with the beam terms EI/$l^3$.

  $[k] = [\,AE/l\ 0\ 0\ -AE/l\ 0\ 0\ ;\ 0\ 12EI/l^3\ 6EI/l^2\ 0\ -12EI/l^3\ 6EI/l^2\ ;\ \ldots\ ]$

- A transformation matrix [T] is used to convert local to global axes for inclined members.

## 9. Derivation of [k] Term by Term (Detailed Steps)

- Start from the [B] matrix. Write the products term by term and integrate from 0 to l:

  $k_{11} = EI \int_0^l (-6/l^2 + 12x/l^3)^2\, dx = 12EI/l^3$

  $k_{12} = EI \int_0^l (-6/l^2 + 12x/l^3)(-4/l + 6x/l^2)\, dx = 6EI/l^2$

  $k_{22} = EI \int_0^l (-4/l + 6x/l^2)^2\, dx = 4EI/l$

  $k_{24} = EI \int_0^l (-4/l + 6x/l^2)(-2/l + 6x/l^2)\, dx = 2EI/l$

- Other terms follow from symmetry $k_{ij} = k_{ji}$ and from equilibrium of the element (sum of forces = 0, sum of moments = 0).
- Check by equilibrium: for column 1 the forces are 12, 6l, -12, 6l (× EI/$l^3$); sum of vertical forces = 12 - 12 = 0, and moment about node

$2 = 6l + 6l - 12\,l = 0$ □.

## 10. Solved Example – Cantilever Beam (Single Element)

- Problem: Cantilever of length $L = 2$ m, $EI = 2 \times 10^4$ kN·m², tip load $P = 10$ kN. Find the tip deflection and slope.
- One element, node 1 fixed ($v_1 = \theta_1 = 0$), node 2 free, load $F_2 = -10$ kN, $M_2 = 0$.

  Reduced equation: $(EI/L^3)\,[12 \;\; -6L ; -6L \;\; 4L^2]\,\{v_2 \;\; \theta_2\}^T = \{-P \;\; 0\}^T$

  From 2nd row: $\theta_2 = 6L\,v_2 / 4L^2 = 1.5\,v_2 / L$

  From 1st row: $(EI/L^3)(12\,v_2 - 9\,v_2) = -P \rightarrow v_2 = -P\,L^3 / 3EI$

  $v_2 = -10 \times 8 / (3 \times 20000) = -1.333 \times 10^{-3}$ m (downward)

  $\theta_2 = 1.5\,v_2/L = -PL^2/2EI = -1.0 \times 10^{-3}$ rad
- Both values match the exact beam theory results $\delta = PL^3/3EI$ and $\theta = PL^2/2EI$, because the cubic shape function is exact for a beam with end loads only.
- Reactions: $\{F_1 \; M_1\} = [k]\{q\}$ gives $F_1 = +10$ kN, $M_1 = +20$ kN·m □.

## 11. Solved Example – Fixed Beam with Central Load

- Problem: Beam of span $L$, both ends fixed, central point load $P$, $EI$ constant. Use two elements with a node at the middle.
- Unknown dof only at the middle node: $v_2$, $\theta_2$. By symmetry $\theta_2 = 0$.

  Equation (middle deflection): $2 \times 12EI/L_e^3 \times v_2 = -P$, $L_e = L/2$

  $(24\,EI \times 8 / L^3)\,v_2 = -P \rightarrow v_2 = -PL^3 / 192EI$
- This agrees with the exact result for the fixed beam.

- End moments from element 1: $M_1 = (EI/L_e^2) \times 6 \times v_2 = -PL/8$ (hogging).

Fixed-end moment for central load = PL/8; deflection under load = $PL^3/192EI$.

## 12. Solved Example – Simply Supported Beam with UDL

- Problem: A beam of length L carries UDL w. Use one element, supports at both ends, $v_1 = v_2 = 0$.

  Reduced equation (rows 2 and 4): $(EI/L)\,[\,4\ 2\,;\,2\ 4\,]\,\{\theta_1\ \theta_2\}^T = \{\,wL^2/12\ -wL^2/12\,\}^T$

- Add the two equations and use symmetry $\theta_2 = -\theta_1$:

  $(EI/L)(4 - 2)\,\theta_1 = wL^2/12 \rightarrow \theta_1 = wL^3/24EI$

- This is the exact slope at a simply supported end under UDL.
- Mid-span deflection from the shape functions ($N_2 = L/8$, $N_4 = -L/8$ at $x = L/2$):

  $v(L/2) = (L/8)\theta_1 - (L/8)\theta_2 = L\,\theta_1/4 = wL^4/96EI = 0.0104\ wL^4/EI$

- The exact value is $5wL^4/384EI = 0.0130\ wL^4/EI$, so one element is about 20% low at mid-span; more elements give better accuracy.

Note: the nodal values are exact for beams, but the deflection between nodes depends on the mesh – use more elements for loads between nodes.

## 13. Stress Recovery in Beam Elements

- Bending stress in an element at distance y from the neutral axis:

  $\sigma = -E\, y\, d^2v/dx^2 = -E\, y\, [B]\{q\}$

- [B] varies linearly along the element, so moment varies linearly: $M(x) = EI\,[B]\{q\}$.
- The nodal end forces give the shear force and bending moment at the nodes:

  $V = F_1$ (at node 1), $M = M_1$

  $\sigma_{max} = M\, y_{max} / I = M / Z$ ($Z$ = section modulus)

## 14. Beam on Elastic Foundation (Short Note)

- When the beam rests on an elastic support (soil), a foundation stiffness matrix is added to the beam [k]:

  $[k_f] = (k_s\, b\, l / 420)\, [156\ \ 22l\ \ 54\ \ -13l\ ;\ 22l\ \ 4l^2\ \ 13l\ \ -3l^2\ ;\ 54\ \ 13l\ \ 156\ \ -22l\ ;\ -13l\ \ -3l^2\ \ -22l\ \ 4l^2\ ]$

- $k_s$ = subgrade modulus, $b$ = width of beam. Total $[k] = [k_{beam}] + [k_f]$.

## 15. Quick Revision Points

- Beam element: 2 nodes, 4 dof, cubic displacement, $C^1$ continuity.
- [k] is 4×4, symmetric and singular; each node has v and θ.
- Equivalent nodal loads for UDL: $wl/2$ and $wl^2/12$.
- Continuous beam: assemble element matrices, apply v = 0 at supports, solve for slopes.
- Exact results are obtained for nodal values with cubic elements when

loads are at nodes.

Important questions: (1) derive shape functions and stiffness matrix for a beam element, (2) analyse a continuous beam (2 or 3 span) with UDL and point load, (3) write the equivalent nodal loads for UDL.